

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

he proposed solution aims to automate the credit card approval process using machine learning, enabling banks and financial institutions to make faster, more accurate, and consistent approval decisions. The system analyzes applicant financial and demographic information, predicts approval or rejection in real time, and minimizes manual effort through an intelligent Flask-based web application deployed on IBM Cloud.

Project Overview	
Objective	The primary objective is to develop an intelligent credit card approval prediction system that uses machine learning algorithms to accurately determine whether an applicant is eligible for a credit card, thereby improving decision-making efficiency and reducing manual processing time.
Scope	The project focuses on collecting and preprocessing applicant data, performing exploratory data analysis, training multiple classification models, selecting the best-performing model, integrating it into a Flask web application, and deploying the solution on IBM Cloud for real-time predictions.

Problem Statement	
Description	Banks receive thousands of credit card applications every day. Manual evaluation is time-consuming, prone to human error, and may lead to inconsistent approval decisions. An automated system is required to improve accuracy, efficiency, and reliability in the credit approval process.
Impact	Implementing this solution will reduce processing time, improve prediction accuracy, assist bank credit analysts in decision-making, minimize operational costs, and enhance customer satisfaction through faster approval decisions.

Proposed Solution	
Approach	Develop a machine learning-based credit card approval prediction system by preprocessing applicant data, engineering relevant features, and training multiple classification algorithms including Logistic Regression, Decision Tree, Random Forest, and XGBoost. The best-performing model will be integrated into a Flask web application for real-time prediction and deployed on IBM Cloud.
Key Features	- Machine learning-based credit card approval prediction. - Real-time approval or rejection through a Flask web application. - Data preprocessing and feature engineering pipeline. - Comparison of multiple classification algorithms. - Interactive and user-friendly web interface. - IBM Cloud deployment for scalable access.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i5 (or higher) CPU
	Memory	8 GB RAM
	Storage	20 GB Free Disk Space
Software	Framework	Flask
	Libraries	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data	Source, Size, Format	Credit Card Approval Dataset (CSV), Application & Credit Records